



Efficiency and spatial resolution of the CASCADE thermal neutron detector



M. Köhli^{a,*}, F. Allmendinger^a, W. Häußler^b, T. Schröder^b, M. Klein^c, M. Meven^d, U. Schmidt^a

^a Physikalisches Institut, Heidelberg University, Im Neuenheimer Feld 226, 69120 Heidelberg, Germany

^b Heinz Maier-Leibnitz Zentrum (MLZ) and Physik Department, Technische Universität München, Lichtenbergstr. 1, 85748 Garching, Germany

^c CDT CASCADE Detector Technologies GmbH, Hans-Bunte-Str. 8–10, 69123 Heidelberg, Germany

^d RWTH Aachen, Institut für Kristallographie und Jülich Centre for Neutron Science at Heinz Maier-Leibnitz Zentrum (MLZ), Lichtenbergstraße 1, 85748 Garching, Germany

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ABSTRACT

We report on the CASCADE project – a detection system, which has been designed for the purposes of neutron Spin Echo spectroscopy and which is continuously further developed and adapted to various applications. It features 2D spatially resolved detection of thermal neutrons at high rates. The CASCADE detector is composed of a stack of solid ¹⁰B coated Gas Electron Multiplier foils, which serve both as a neutron converter and as an amplifier for the primary ionization deposited in the standard counting gas environment. This multi-layer setup efficiently increases the detection efficiency and by extracting the signal of the charge traversing the stack the conversion layer can be identified allowing a precise determination of the time-of-flight. The spatial resolution is found by optical contrast determination to be $\sigma = (1.39 \pm 0.05)$ mm and by divergence corrected aperture measurements $\sigma = (1.454 \pm 0.007)$ mm, which is in agreement with the simulated detector model. Furthermore this enabled to investigate and describe the non-Gaussian resolution function. At the HEiDi diffractometer the absolute detection efficiency has been studied. At 0.6 Å for the 6 layer detector, which is currently part of the RESEDA spectrometer, an efficiency of 7.8% has been measured, which by means of Monte Carlo simulations translates to $(21.0 \pm 1.5)\%$ for thermal neutrons at 1.8 Å and $(46.9 \pm 3.3)\%$ at 5.4 Å.

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1. Introduction

The world of detectors used in neutron scattering instrumentation has changed. Much of what once was established has been discarded. By alert [1] on the future helium-3 supply [2], which is critical to perspectives of the European Spallation Source [3], the run on substitutional technologies started [4,5]. Most of the solutions could be adapted from developments of particle physics. The CASCADE neutron detection unit [6] is such a system, which was designed specifically for the purposes of Neutron Spin Echo (NSE) spectroscopy [7,8]. This method and its successors, Neutron Resonance Spin Echo (NRSE) [9] and Modulation of Intensity by Zero Effort (MIEZE) [10,11], demanded a highly granular and time resolved detector to be operated efficiently at high rates. The CASCADE design is based on a combination of solid boron-10 coatings in several layers, modern gas amplification stages, a

microstructured readout, multichannel ASICs (Application Specific Integrated Circuit) and FPGA (Field Programmable Gate Array) hardware triggered data acquisition. It is successfully in operation at the Forschungs-Neutronenquelle Heinz Maier-Leibnitz (FRM II) at the instruments RESEDA [12,13] and MIRA [14,15]. This paper presents the characterization of the 6-layer detector at RESEDA.

2. Neutron detection by solid converters

Thermal neutrons ($\lambda = 1.8$ Å, $E_{\text{kin}} = 25.2$ meV) cannot be detected directly in an efficient way as they are in equilibrium with the environment and so the change of entropy in the sensitive active medium required for a significant signal to noise ratio is too small. Converters are used to generate a strong signature by processes which are specifically sensitive to thermal neutrons such as nuclear absorption followed by immediate de-excitation [16]. Solid boron requires a hybrid detector with both stages separated. The particles emitted after absorption are charged core fragments of

* Corresponding author.

E-mail address: koehli@physi.uni-heidelberg.de (M. Köhli).

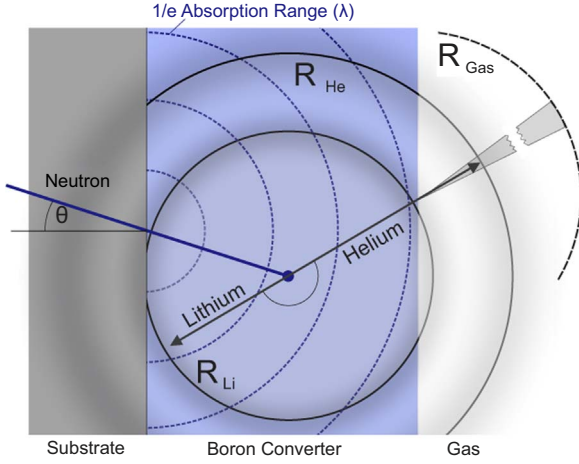


Fig. 1. Physics of a hybrid boron converter layer: a neutron impinges at an angle θ normal to the surface. The absorption probability scales wavelength dependent by the Beer-Lambert law. By the absorption process lithium (Li) and helium (He) ions with fixed energies are created by fission. In the converter medium they lose energy by collisions leading to a Bragg distributed range, which is different for both agents. After leaving the solid layer an ionization track is produced according to the remaining energy at the boundary.

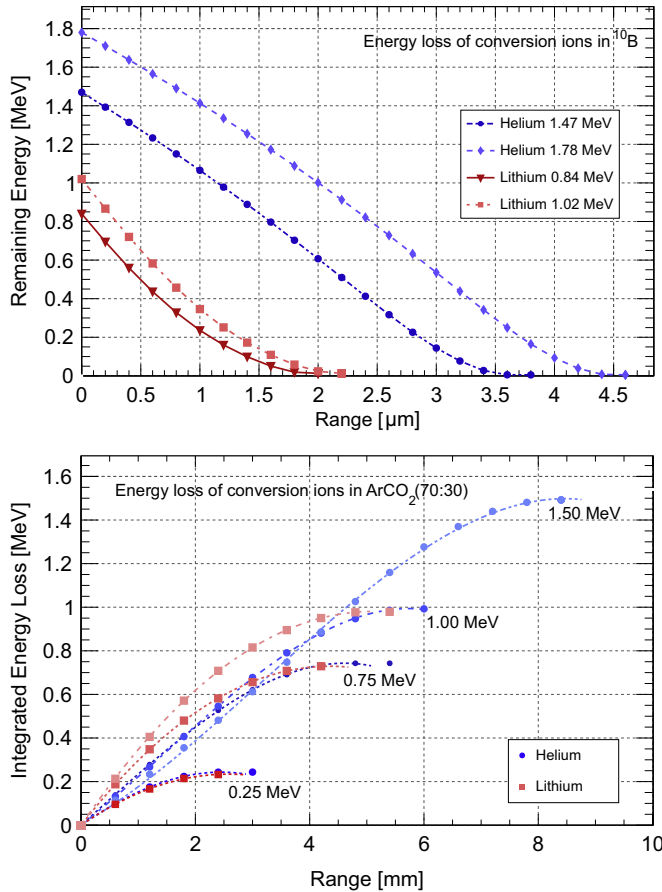
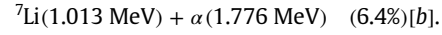
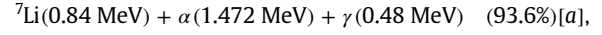
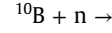


Fig. 2. Calculated energy loss of helium and lithium ions, whereas the initial kinetic energy is given by the neutron conversion reaction. (Top) Energy-range relation shown for solid ^{10}B and a density of $13.05 \cdot 10^{22}$ atoms/cm 3 with 4th order (helium) and 3rd order (lithium) polynomial fit functions. (Bottom) After leaving the conversion layer the energy spectrum of the ions is continuous, an exemplary set of energy-range relations in Ar:CO $_2$ with 3rd order (helium) and 2nd order (lithium) polynomial functions are shown.

high stopping power, thus converters have to be as thin as micrometers. The system in total can be understood and optimized by means of analytical modeling [17] or Monte Carlo transport simulations [18]. In the following the processes in thin films will be discussed, detailed descriptions for specific geometries can also be found in [19,20], whereas for now a simple single layer like Fig. 1 is considered. The capture cross section of boron at $\lambda = 1.8$ Å is (3844 ± 3) barn [21] leading to an attenuation length of $20 \mu\text{m}$. Two absorption reactions occur:



In the boron medium itself the particles are emitted isotropically and back to back. They lose kinetic energy mainly by ionization. The average maximum ranges are from $(1.69 - 1.90) \mu\text{m}$ for lithium to $(3.27 - 4.05) \mu\text{m}$ for helium ions in the branches [a] and [b] with a straggling variation in the order of $\sigma \approx 5\%$ according to calculations by SRIM [22]. These values, compared to bulk density, are similar for boron carbide and boron in natural isotope concentration and limit the active thickness of each layer. Fig. 2 shows the energy-range-relation of the conversion products which are depicted in Fig. 1 as spheres. Two geometries are possible: (1) the front side irradiation describes the case of the active surface being oriented towards the incoming neutrons and therefore the detection medium being situated prior in beam direction. (2) Vice versa the backside irradiation refers to the inverted geometry. In that case and for multi-layer detectors the amount of inactive converter material decreases the efficiency. After leaving the boron layer the ion spectrum is continuous due to the interaction length distribution in the solid, which is a function of conversion depth of the neutron and escape angle of the conversion pair. Fig. 3 provides an overview about the shape of the spectra for different converter layer thicknesses in front-side irradiation. The maximum kinetic energy of both decay branches can be kept by a neutron conversion just below the boron surface. Absorption reactions more deep in the material lead to a substantial loss of energy and therefore extend the spectrum to the left with increasing path length. It is also shown that due to the lower initial energy and higher stopping power the lithium ions make up the lower end of the spectrum. Already at comparatively thin layers these ions tend to be captured within the boron. For a layer thickness beyond $1.2 \mu\text{m}$ a low energy pile-up appears leading to stop the further contribution of the lithium ions to the detectable signal. As either a noise or gamma rejection threshold has to be applied, it has to be considered in the efficiency budget that for thick layers the fraction of ionizing particles below threshold increases significantly.

3. The CASCADE detector concept

The CASCADE detector is based on thin films of 99% enriched boron-10. The conversion products create a signal by ionization in the standard counting gas Argon:CO $_2$ of mixtures from (70:30) to (88:12). To overcome the limited efficiency of a single film of (5–7)% at 1.8 Å a stack of several layers is required. The detector, see Fig. 4, is comprised of GEMs [23], which act as a substrate for the converter as well as they are charge transparent – at $180 \mu\text{m}$ pitch the CERN-GEM has an optical opacity of $\approx 88\%$. GEMs being amplification stages themselves can compensate for the charge lost on the surface by a high gas gain in the holes. The technological concept of such an assembly in general and a variety of applications which are based on GEMs is presented in [24] and, especially in the field of neutron detectors, realizations are described in [25–

27]. Such has been realized as well with the previous version of the CASCADE detector, for which a detailed technical overview is provided in [6]. The following description explicitly focuses on the additions to the 2010 detector assembly. It is now further possible

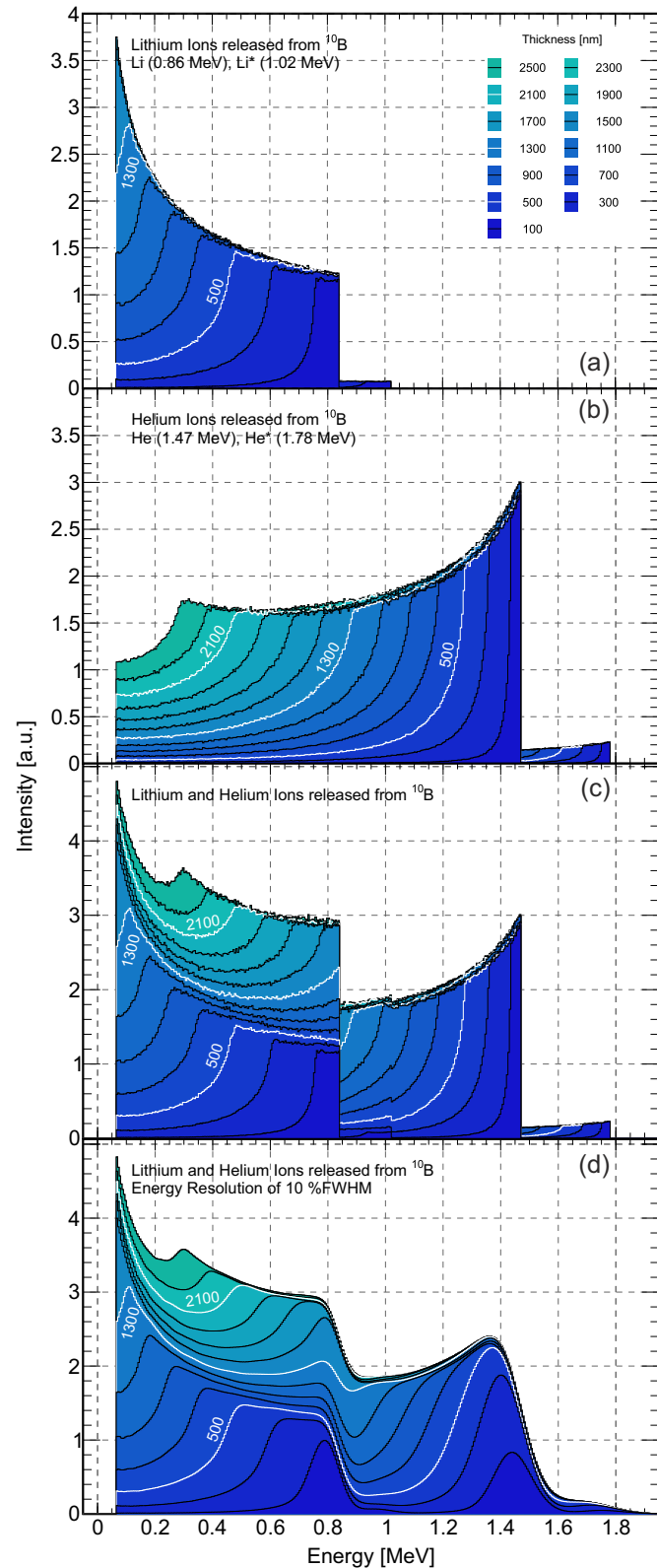


Fig. 3. Simulated kinetic energy distributions of the conversion products lithium (a) and helium (b) and their spectrum in total (c) after leaving solid front side irradiated ^{10}B layers of various thicknesses at a density of 2.18 g/cm^3 .

to detect the positive signal on the copper coated GEM electrode surface in addition to the negative electron signal on the micro-structured readout. This allows to identify the origin of the neutron conversion as the charge can be tracked while traversing through the stack (for an example of the Spin Echo mode see [28]). Grids are inserted in-between each stage in order to decouple the GEMs electrically and to limit the generated charge. The ideal gain factor in each stage, which is a result of amplification in the holes, charge carrier collection and extraction efficiency determined by the electric field ratios, would be unity. In order to maximize the z-channel sensitivity the GEMs have to be operated at an effective gain of approximately 4. Therefore a metalized mesh with 33% open area is chosen [29]. A good signal to crosstalk ratio can be achieved by 2 nF readout and 10 nF decouple capacity. The CASCADE detector described herein has 6 layers divided into two mirrored half-spaces. Whereas layer 1 and 6 are boron coated drift electrodes followed by a GEM, the in-between stages are single sided boron coated GEMs. The individual thickness of the layers according to the specifications by the manufacturer (CDT GmbH) from top to bottom are: $1.5 \mu\text{m}$, $0.8 \mu\text{m}$, $0.99 \mu\text{m}$, $0.95 \mu\text{m}$, $0.8 \mu\text{m}$, and $2.0 \mu\text{m}$. Functionally there is a distinction between the imaging mode (2D) and the TOF mode (3D). In the latter one besides the spatial event reconstruction on the readout a correct z-layer identification is required. Due to the high readout capacity of the whole GEM the sensitivity on the z-channel is lower, therefore tracks with a small energy deposition have a significantly weaker signal strength. This results in a higher effective detection threshold. According to the simulations presented in Fig. 3 for thick layers a pile-up at the lower end of the energy deposition spectrum emerges. After this measurement campaign the original drift electrode coated by $2 \mu\text{m}$ of boron has been replaced by a $1 \mu\text{m}$ coating. The distances between the (boron) layers are (2.4–4.8) mm, see Fig. 4. This allows for applying a stopping power discrimination against gamma background. As the mean free path of photons or their direct products in the gas in most cases is shorter than the conversion particle range the quantity measured

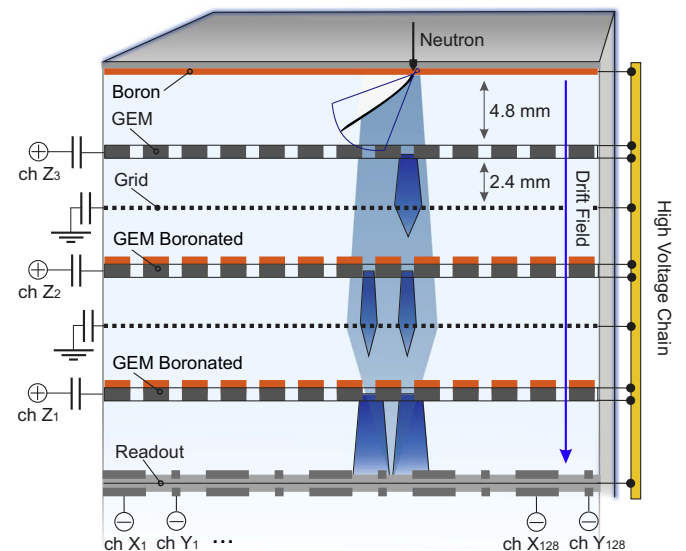


Fig. 4. Schematic of the CASCADE detector of one half space: Thermal neutrons are converted in boron layers (orange), which are either coated on the aluminum casing (top) or on GEMs (below). The conversion products, lithium or helium, which leave the boron, deposit a fraction of the total kinetic energy ionizing the counting gas. The semi-transparent GEMs compensate for the charge loss by gas amplification in strong electric fields in the holes. The electrons are projected towards a double sided readout with crossed stripes and additionally the signal induced on the bottom side of the GEM electrodes is measured. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

is the energy loss dE/dx . The gamma rejection is facilitated as the stopping power of ions is by orders of magnitude larger compared to electrons. In comparison to Fig. 3 showing the remaining kinetic energy of the conversion products after leaving the boron the actual energy depositions in the 6 layer detector according to the real geometry are presented in Fig. 5. In order to relate this distribution to the gamma sensitivity, the continuous slowing down approximation (CSDA) range of electrons of for example 20 keV in Ar:CO₂ is 5 cm [30], therefore background energy depositions larger than 10 keV are strongly suppressed. The double sided x,y readout board consists of crossed stripes in a nested comb structure with a pitch of 1.56 mm. At (20×20) cm² active surface the detector has 128×128 pixels, see also [31]. The backplane electronics is based on the 64 channel charge sensitive amplifier CiPix [32,33], a modified version of the HELIX128 ASIC [34], which is operated at 10 MHz leading to a time stamp resolution of 100 ns. One additional CiPix is necessary to read out the charge signal induced on the GEM surfaces (z-channel). The digitized signals are handled by a Virtex-2 FPGA, which can either produce a zero-suppressed raw data output or it can analyze the data patterns by firmware algorithms in the triggered event mode, which is necessary for example in combination with a phase locked loop Spin Echo operation. For $8 \times$ oversampling a MIEZE frequency of up to 1.25 MHz can be achieved, whereas the standard $16 \times$ scanning is suitable for 625 kHz.

4. Detector simulation

The detector was simulated by the Monte Carlo neutron transport code URANOS (Ultra Rapid Neutron Only Simulation) [35]. The software was especially tailored to the specific needs of a hybrid time resolved detector. It includes neutron physics, following the implementation described by OpenMC [36] and linked to the cross sections database ENDF/B-VII.1 [21] and JENDL-4.0 [37], an analytic model of charged particle transport in solids and gases derived from SRIM and a simplified gas amplification model for the GEMs and the geometric layout of the readout structure (see also Fig. 2). This allows for a ray tracing based time resolved analysis of the full signal generation path. Fig. 6 shows the calculation scheme. For efficiency measurements it is especially important to consider the scattering of the neutrons in the detector material itself as the polyimide materials used, for example in GEMs, contain a non negligible amount of hydrogen. For characterizing the system this leads to a non-uniformly broadened resolution function which is different for each layer and which changes on the percent to sub-percent level according to the energy and angle distribution of the incoming beam.

5. Spatial resolution

The spatial resolution was determined at the RESEDA cold neutron spectrometer at the FRM II of the Technische Universität München, where the detector was placed at the end of the guide after a Dornier velocity selector providing neutrons of 5.4 Å (11% FWHM). The instrument equipped with the CASCADE detector is designed for the purposes of neutron Spin Echo spectroscopy, especially the MIEZE technique, which requires a high spatial and time resolution. Two approaches were carried out, (1) the edge multisampling method and (2) the Siemens star characterization, and both are compared to results from the detector simulation. The spatial resolution is nontrivial to describe, as it is shown in Fig. 7 and analyzed later. Therefore, two different methods are used to account for the non-Gaussian point spread function (PSF). In this article we are using the 1σ standard deviation for the

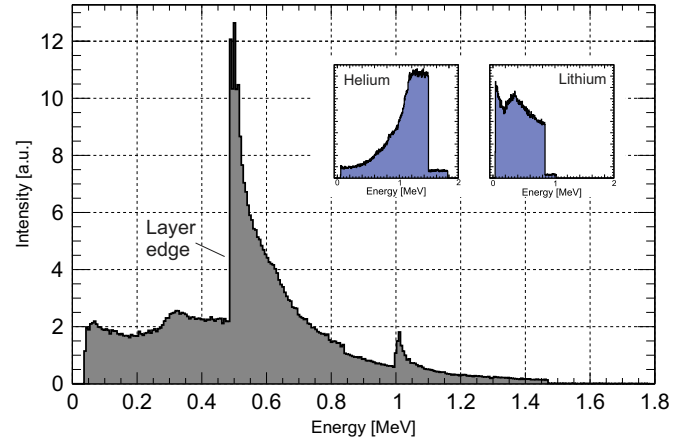


Fig. 5. Simulation of the gas energy deposition of the conversion products for the CASCADE detector. The inlays show the energies of the particles escaping the boron coatings. The geometry of each layer limits the track length and therefore the available signal energy. As different layer thicknesses are present in the detector, the distributions, which are added up in this graph, are not uniform.

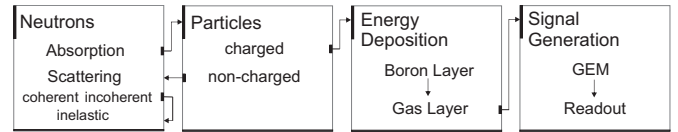


Fig. 6. Calculation schematic of URANOS. The path includes the individual segments neutron physics, particle generation after absorption or absorption-like processes, energy loss and deposition calculation and finally the projection onto the readout through the amplification stages.

spatial resolution description as the full width at half maximum underestimates the width of the PSF due to the very specific shape. For the optical **Siemens star method** a gadolinium coated pseudo random chopper was directly placed in front of the detector. Its sequence with the smallest sections having 2.835° is described in [38]. A graphite sample provides a homogenous illumination. This contrast measurement is shown in Fig. 8. The resolution σ_{opt} is calculated by determining the smallest possible spatial structure to be mapped by measuring the radius r at which the contrast is lost. For a result of $r = (56 \pm 2)$ mm we find

$$\sigma_{\text{opt}} = \pi r \Delta\phi / 360^\circ \rightarrow \sigma_{\text{opt}} = (1.39 \pm 0.05) \text{ mm} \quad (1)$$

For the **edge multisampling method** a straight cut cadmium absorber was placed in front of the detector at various distances oriented in a flat inclination angle with respect to the readout structure, see Fig. 9. By subtracting a bare illuminated shot without absorber the edge spread function f_{es} is determined. It is a convolution of the Heaviside step function $\theta(l(x, y))$ along a straight line $l(x, y)$ and a Gaussian point spread function $f(x, y, \sigma)$:

$$f_{\text{es}} = \int_{-\infty}^{\infty} \theta(l(x, y) - r) f(r, \sigma) dr, \quad (2)$$

whereas the standard deviation σ for an intrinsic detector resolution σ_0 and a beam with divergence angle α in a distance d is calculated by:

$$\sigma = \sqrt{\sigma_0^2 + \tan(\alpha)^2 d^2} \quad (3)$$

This method involves three elements: (1) The detector resolution can be determined with high statistics over the whole active surface, (2) pixel size effects do not have an influence as the readout is sampled randomly by the aperture at different fractions of the pixel area and (3) the beam divergence spread will be

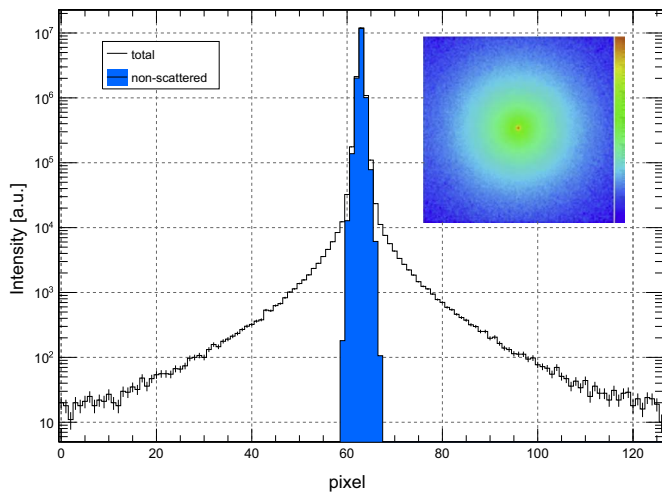


Fig. 7. Point spread function of the detector visualized by a profile cut of the simulation of a pencil shaped beam. Those neutrons which did not undergo scattering in the detector itself (blue) are separated from the total counts. The inset shows the full active area in 2D. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

removed from the measured resolution by extrapolating to a zero distance from the detector to the edge.

The result from the fit for the intrinsic detector resolution is $\sigma_0 = (1.454 \pm 0.007)$ mm, see also Fig. 10. This includes three effects. Firstly, the resolution of the readout can be denoted as $\sigma_r = 1.56 \text{ mm}/\sqrt{12} = 0.45 \text{ mm}$.¹ Secondly, the ionization track lengths of the conversion ions in the gas make up the largest part of the error budget. And thirdly there is a broadening of the resolution function by the elastic scattering of the neutrons on the hydrogen in the detector material, which can be observed in Fig. 9 as a corona-like effect below the cadmium absorber.

Both aspects are addressed by Monte Carlo simulations. Beginning and end of an ionization track cannot be separated in the detector due to the high electron drift velocities of approximately 3 cm per μs at 1 kV per cm [39]. Therefore the spatial resolution (1σ) corresponds to 1/2 of the projected track length. This mean free path varies by the size of the gap between the layers and also depends on the type of conversion product as the lithium ions escape the boron with lower energies compared to helium. Fig. 11 shows the track length distributions. The result for the total standard deviation is $1\sigma = 1.817 \text{ mm}/2 = 0.908 \text{ mm}$. Separated into conversion product types the results are $\sigma_{\text{He}} = 2.24 \text{ mm}$ for helium and $\sigma_{\text{Li}} = 1.31 \text{ mm}$ for lithium ions. The active detection volume of the detector contains hydrogen based materials. The main contributors to the scattering cross section are the GEMs (50 μm), the readout structure (100 μm) and the thermoplastic grids (44 μm). Instead of a gamma background this effect produces a homogeneous low level illumination by delocalized neutrons. As the scaling of scattering and absorption cross sections by wavelength is different the impact changes with the neutron energy. Most of the scattered neutrons are absorbed very closely to their origin. Typically the amount of neutrons with a dislocation larger than one track length standard deviation stays well below 0.5%, although at 5.4 Å for example 19% of the detected neutrons have undergone at least one scattering process. The total point spread function of a simulated pencil shaped neutron beam at 5.4 Å is shown in Fig. 7. The self scattering background plateau stays at least 4 orders of magnitude below the signal. In this simulation the total error defined by the 68.3% quantile is found to be 1.56 mm. The error for self scattering effects then is calculated by

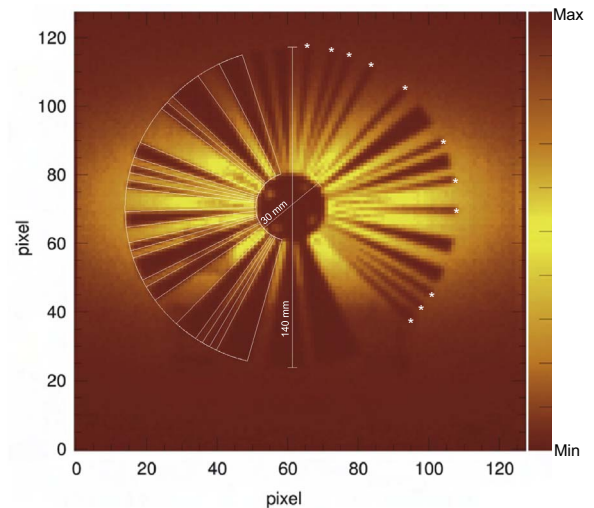


Fig. 8. Image of a pseudo random chopper illuminated by a cold neutron beam scattering off a graphite characterization sample at RESEDA. The segments marked by * cover an opening angle of $\Delta\phi = 2.835^\circ$. The linear two-fold color code is applied to emphasize the contrast. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

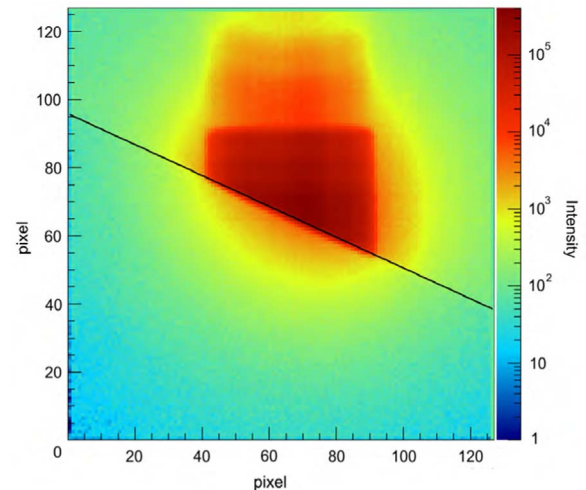


Fig. 9. Image of a cold neutron beam at RESEDA in logarithmic scaling with a cadmium aperture in front for determining the spatial resolution. The black line indicates the fit result of the edge spread function.

subtracting the known errors from the total budget.

This error budget for the spatial resolution is denoted in Table 1. The specific shape of the point spread function with its non-Gaussian core and long tails can be targeted in different ways. In the optical measurement the aperture function sits on a background of constant illumination. Here mainly the short range distribution contributes as the contrast loss of two edges approaching is determined and therefore it leads to the smallest result for the spatial resolution. By fitting the edge spread function to the projection of the aperture the effect of self scattering becomes more important, which is visible by the corona feature in the logarithmic intensity display of Fig. 9. Whereas in the measurements presented we cannot strictly distinguish between environmental self scattered background in the simulation we have tracked each neutron individually, which enables to include the plateau of the point spread function. This leads to the result, that the homogeneously spread outliers contribute to an increase of the standard deviation.

¹ The standard deviation of a step function is pitch divided by $\sqrt{12}$.

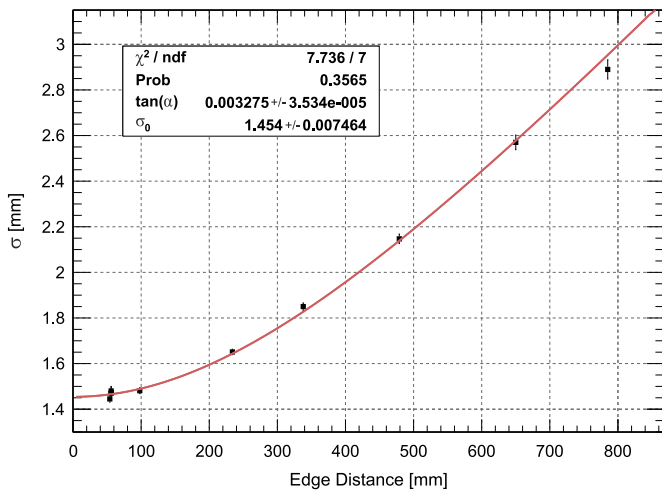


Fig. 10. Determination of the spatial resolution. Fit results for the edge spread function for increasing distances between detector and aperture.

6. Detection efficiency

The detection efficiency was measured at the HEiDi single crystal diffractometer [40] at the FRM II, which provides monochromatic hot neutrons. The neutrons from the reactor are wavelength selected by a monochromator. A 0.5 mm erbium foil removes a band in the hot spectrum below 0.5 Å [41]. Furthermore the contamination is reduced by diffracting the beam by a cubic single crystal of silicon (125 mm³) in Bragg condition at the sample position. Fig. 12 shows the spectrum at the hot source at the beginning of the guide of the HEiDi instrument. As the flux density of high energetic neutrons below the measurement range is reasonably high a beam contamination by integer fractions n of λ can occur. Small wavelength contributions are absent due to the product of spectral density and reflectivity of the sample. For Ge(311) the second reflex (622) is forbidden, which as well holds for Ge (533). The second reflex of Ge(422) and the third of Ge(311) are filtered by the erbium foil. The according domains are also depicted in Fig. 12. In total the contribution of fractions of the selected wavelength stay below 0.5%. In addition a 15' collimator is used to reduce the horizontal divergence. Table 2 gives an overview about the configuration. The standard HEiDi detector is an

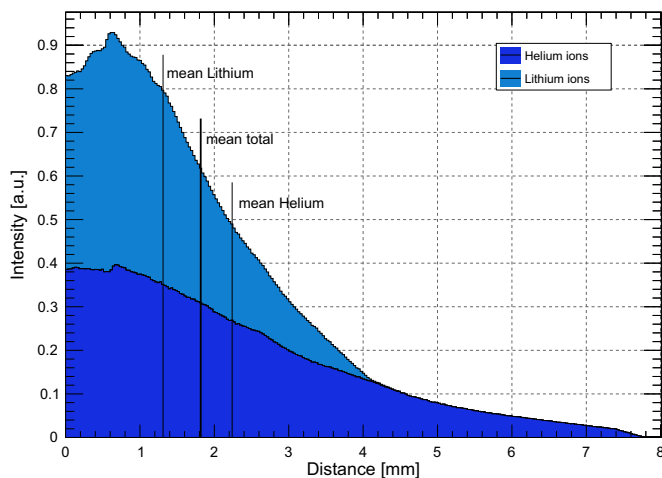


Fig. 11. Track length distribution of helium (blue) and lithium (red) conversion ions in the detector. The mean corresponding to 2σ is indicated by the vertical line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

Table 1

Spatial resolution error budget for the 1σ standard deviation. The total error is determined by measurement, the pixel effect by geometry, the ion smearing and the self scattering are determined by simulation. The contribution ratios refer to the total variance.

Resolution	Result (mm)	Contribution
Pixel size	0.45	10%
Ion smearing	0.908	39%
Self scattering	1.04	51%
Total	(1.454 ± 0.007)	
Siemens star	(1.39 ± 0.05)	

Eurisys 73NH17/5X ³He counter tube with a pressure of 5 bar, a 4 mm Alumina entrance window and an active length of 170 mm at a diameter of 50 mm. The absorption of the entrance window is specified as 2% for thermal neutrons, which translates to 1.3% for the used wavelength of 1.17 Å. The tube can theoretically absorb more than 99.9% of the remainder. However, since no measurements exist, the range provided by the manufacturer is used. In front of the detector at a distance of 200 mm a shielded aperture of 22 mm × 28 mm is mounted to limit the viewing angle towards the sample. Table 3 summarizes the reference measurements. The efficiency measurements were done by placing the CASCADE detector at the position of the reference counter with the full active area oriented towards the sample, therefore the shielding covered only the backward facing 2π . All measurement runs were conducted as a function of the inclination angle to investigate at the same time the characteristics of the signal distribution in each conversion layer. In the analysis it was found that by omitting to insert the vertical collimator before the sample the beam broadened to a large extent. In a comparative overview the spots for each wavelength as measured by the CASCADE detector are shown in Fig. 13. Additionally the beam at 1.17 Å showed an angular deviation of $(0.91 \pm 0.15)^\circ$ as against both other wavelength runs, which can be a consequence of a slightly different focusing at each monochromator orientation. At a distance of 480 mm at the position of the aperture this translates to approximately 1 mm. We assume that this lead to a small fraction of the beam being absorbed and undetected by the reference counter. Therefore the efficiency calculation is slightly overestimated as the ³He count rate is compared to the results from the CASCADE detector, which does not have an artificial limitation on the viewing angle. Table 4 summarizes quantitatively the results of the efficiency measurements in comparison to the values obtained from the detector simulation. Fig. 14 shows the full angular data. For the efficiency analysis a circle cutout of 35 mm centered around beam spot has been chosen to accurately extrapolate the inhomogeneous background from the complementary pixel set. Therefore in the simulation two values are given: one corresponding to the full detection efficiency and one for the expected count rate in the selection region. This distinction is attributed to the special shape of the spatial resolution function, see Section 5. It has to be noted that in a generalized linear model, independent of the geometry, a scaling of the detection efficiency by λ is expected. The simulation can follow such an approximation for small wavelengths, however, the data points themselves are not strictly consistent with such an hypothesis.

For the reasons discussed before, which lead to deviations of the absolute count rate determination, in this measurement campaign we conservatively set the count rate uncertainty to 7%. A deviation of similar magnitude was also found in the Jalousie campaign [19], which is as well technically different as the efficiency estimation was carried out by an analytical model. In

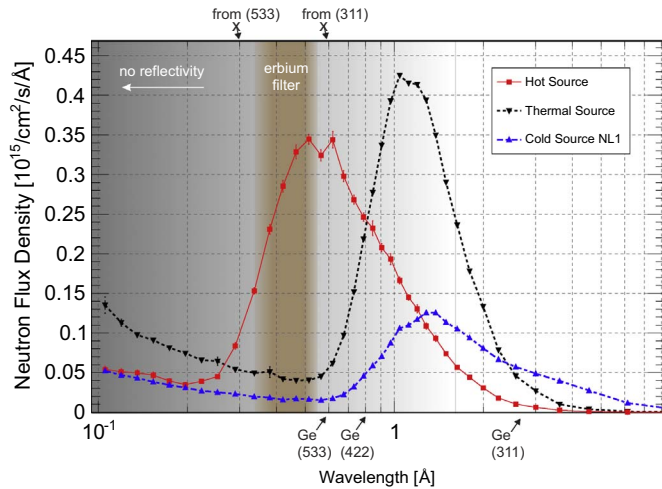


Fig. 12. Neutron flux density as a function of wavelength at the FRM II for the different sources. Simulations published by the FRM and refined in 2010 by A. Röhrmoser.

Table 2
Configurations for the wavelength selection at the HEiDi diffractometer.

λ (Å)	Monochromator	Si (hkl)	2θ
0.593	Ge(533)	(404)	36.00°
0.794	Ge(422)	(404)	48.84°
1.17	Ge(311)	(202)	35.52°

Table 3
Results for the reference measurements of the HEiDi ^3He counter for the specified wavelengths.

λ (Å)	Efficiency (%)	Bragg rate (1/s)	Background (1/s)
0.593	95.5	(512 ± 2)	(3.76 ± 0.18)
0.794	97.0	(2618 ± 2)	(20.0 ± 0.8)
1.17	98.5	(8015 ± 8)	(38.3 ± 0.6)

conclusion we obtain for the efficiency at the selected wavelengths in the imaging mode: $(7.8 \pm 0.5) \%$ at 0.593 Å, $(11.5 \pm 0.8) \%$ at 0.794 Å and $(16.2 \pm 1.1) \%$ at 1.17 Å.

The overall efficiency on the whole range of wavelengths is analyzed by the detector simulation model, which is shown in Fig. 15. We obtain an efficiency for thermal neutrons (1.8 Å) of $(22.7 \pm 1.6) \%$ and $(50.0 \pm 3.5) \%$ for the 5.4 Å of the cold source at RESEDA. The updated detector with the 1 μm boron coating on the backside drift electrode has approximately 93% of the count rate compared to the values determined in the measurements presented here.

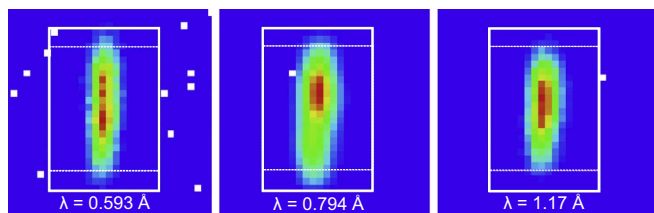


Fig. 13. Spatial representation of the diffracted spots for each wavelength in a linear scaling relative to the maximum value present in the frame. The white boxes show the HEiDi ^3He counter aperture: in solid lines adapted to the beam divergence at the position of the CASCADE detector and dashed as scaled to the actual extent.

Table 4

Results of the efficiency measurements of the 6-layer CASCADE detector in the imaging mode for the specified wavelengths compared to the simulated expectation values.

λ (Å)	Efficiency [%]		
	Measured	Simulation spot	Simulation all
0.593	7.82 ± 0.04	7.95	8.33
0.794	11.47 ± 0.04	10.50	11.00
1.17	16.21 ± 0.04	14.95	15.62

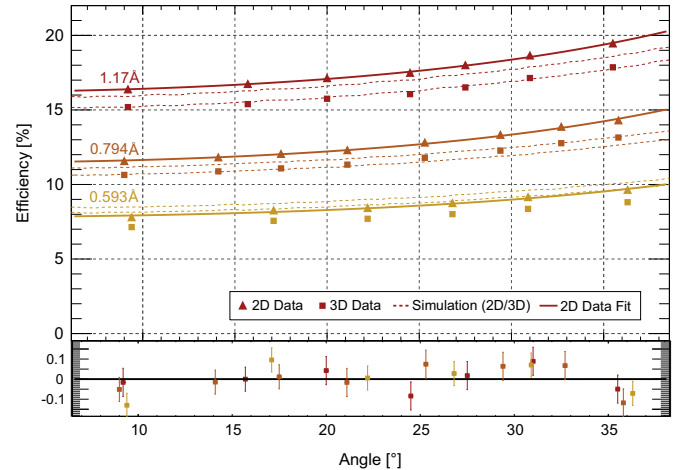


Fig. 14. Efficiency of the CASCADE detector for three different wavelengths as a function of the inclination angle for the imaging and TOF mode. The residuals of the fit of an exponential to the data points are plotted in the lower panel. In comparison the simulation results are shown for the whole active detector area (upper curve) and the spot corresponding to the reference tube aperture (lower curve).

7. Summary and outlook

In this paper the CASCADE 6-layer thermal neutron detector for Spin Echo spectrometry has been characterized by means of spatial resolution and detection efficiency. The point spread function of the detector is not Gaussian due to scattering effects and the track topology of the conversion ions. Therefore the spatial resolution has been determined by two methods. The optical contrast measurement leads to $\sigma = (1.39 \pm 0.05) \text{ mm}$ and the edge multisampling to $\sigma = (1.454 \pm 0.007) \text{ mm}$. At the HEiDi diffractometer the detection efficiency has been measured by monochromatic beams at three different wavelengths. In order to extrapolate the results to any neutron energy by means of Monte Carlo methods the error is estimated by the deviation to the simulated expectation value. This mainly accounts for uncertainties in the experimental setup. The obtained thermal neutron detection efficiency in the current configuration is $(21.0 \pm 1.5) \%$ and $(46.9 \pm 3.3) \%$ at 5.4 Å.

The modular CASCADE design offers further upscaling for two options. The multi-layered assembly allows to increase the effective amount of converter material in the beam to achieve a higher efficiency at a high spatial resolution. Future (MIEZE) Spin Echo systems extending to longer wavelengths and higher frequencies would strictly limit the effective detector depth as the Spin Echo group can be as short as a few millimeters. For this case ^{10}B systems like the design presented are cost efficient to modularly instrument larger areas.

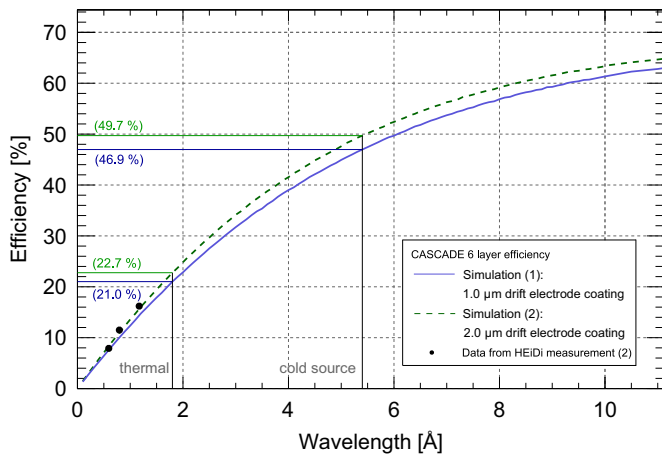


Fig. 15. Efficiency of the CASCADE detector in both drift electrode configurations for the imaging mode at various wavelengths. The HEiDi measurements are extrapolated to higher neutron wavelengths by the detector simulation model.

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